Environmental Condition Monitoring and Decision Making System Using Fuzzy Logic Method

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***Abstract*— Currently, air pollution is still a problem that requires special attention, especially in big cities. Air pollution can come from motor vehicle fumes, factory smoke or other particles. To overcome these problems, a system is made that can monitor environmental conditions in order to know the good and bad of air quality in an environment and is expected to be a solution to reduce air pollution that occurs. The system created will utilize the Wireless Sensor Network (WSN) combined with Waspmote Smart Environment PRO, so that later data will be obtained in the form of temperature, humidity, CO levels and CO2 levels. From the sensor data that has been processed on Waspmote, it will then be used as input for data processing using a fuzzy algorithm. The classification obtained from sensor data processing using fuzzy to monitor environmental conditions there are 5 classifications, namely Very Good, Good, Average, Bad and Dangerous. Later the data that has been collected will be distributed to Meshlium as a gateway and will be stored in the database. The process of sending information between one party to another needs to pay attention to the confidentiality of data and information. The final result of the implementation of this research is that the system is able to classify values using fuzzy algorithms and is able to secure text data that will be sent to the database via Meshlium, and is able to display data sent to the website in real time.**

Keywords— ***Wireless Sensor Network (WSN), Libelium,*** **Cryptography*****, Fuzzy Logic, Advanced Encryption Standard (AES).***

I. INTRODUCTION

Air pollution is a problem in a number of areas in Indonesia, especially in big cities and industrial areas. Air pollution is one of the leading causes of premature death. Air pollution itself occurs when there are foreign materials or substances in the air that cause changes in the composition (composition) of the air from its normal state. The presence of a certain amount or foreign substance in the air and in the air for a long time can interfere with the life of humans, animals and plants. If that happens, then the air has been polluted [1].

The air in urban areas which has a lot of industrial and technological activities as well as dense traffic, the air is relatively not clean anymore. WHO says more than 90 percent of the world's population suffers from air pollution which has drastic effects on health, especially in children [2].

To be able to overcome this problem, an environmental health monitoring system was created using a Wireless Sensor Network (WSN) or wireless sensor network, with the ability to unify environmental conditions in the form of air temperature, humidity, CO levels and CO2 levels. The performance of the sensors that will be used is to collect data and will be distributed to the cloud server and sent to the data center. Based on the information that has been collected in the data center, a decision will be taken to give a warning about the environmental conditions.

When the sensor delivers data to the cloud server, a security approach based on AES will be used to ensure that no information is leaked or modified by others. Data integrity on WSN can be preserved using the cryptographic technique AES (Advanced Encryption Standard). AES works by converting plaintext data into ciphertext in the encryption process and converting ciphertex data into plaintext in the decryption process [3].

II. SYSTEM DESIGN

This section describes the system design and the steps for making the system as a whole which is the main discussion in this final project. At this stage there are several jobs that are carried out, as shown in Figure 1.

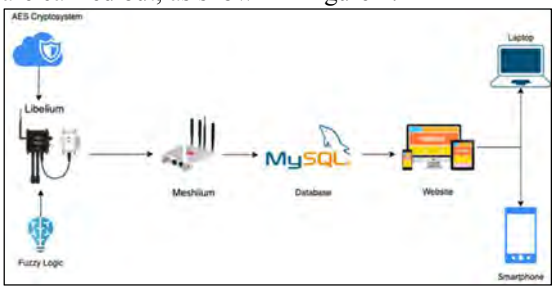

Fig 1. Block Diagram

A. *Wireless Sensor Network (WSN)*

Wireless Sensor Network (WSN) is a collection of sensor nodes that are connected wirelessly (wireless) and work together to monitor a situation over a long period of time with a high level of accuracy. WSN can measure environmental conditions such as temperature, noise, pollution level, humidity, wind, and so on. WSN generally consists of a sensor field, a number of sensor nodes, a sink node, and a data center. The sensor field is the area to be monitored, a number of sensor nodes function as information collectors from the surrounding environment, the sink node is tasked with collecting information obtained from a number of sensor nodes, where the sink node can also function as a cloud server (connection to the external network), while The data center is a processing and storage of information received from the sink node via the internet.

In WSN, each sensor node can individually sense the surrounding environment and cooperate with other sensor nodes to obtain the desired information. Therefore, the purpose of the wireless sensor node is to get an overview of the environment through the sensor and wirelessly communicate to get an image and help other nodes to provide an overview of the environment to be observed [4].

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B. *Fuzzy Logic Method*

Fuzzy logic has a membership degree that has a value range of 0 (zero) to 1 (one). Fuzzy logic is an improvement from Boolean logic, where classical logic is expressed in binary terms (0 or 1, yes or no). Fuzzy logic replaces the Boolean truth level with a truth level, meaning that fuzzy logic has uncertainty concepts such as "a little", "fair", and "very". The implementation of fuzzy logic in the control system is a leap of innovation in the control system. This is because the control using a fuzzy system is more precise when compared to a digital control system which only controls an on/off device. Fuzzy is expressed in degrees of membership and degrees of truth. Therefore, something can be said to be partly right and partly wrong at the same time[5].

C. *Website View*

A website is a page that features text, graphics, audio, and video to present information. In this website, graphs and tables comprising sensor data for temperature, humidity, carbon dioxide (CO2), as well as the conditions of the environment. The website used is a website that is deployed from a server ip, which means that this website can be accessed by admins anywhere and using any device.

1. ***Login Page***, because the website in this final project is designed for admins, there is a login page when accessing the website. On this page, the admin will enter the username and password that have been registered in the database.

Fig 2. Login Page

2. ***Dashboar*d,** the dashboard or main page will display the values of Temperature, Humidity, CO levels and CO2 levels as well as the current conditions of the environment, where in the event of the addition of the latest data, the contents of this display will be updated according to the new data. This dashboard page also displays a graph of CO levels and CO2 levels.

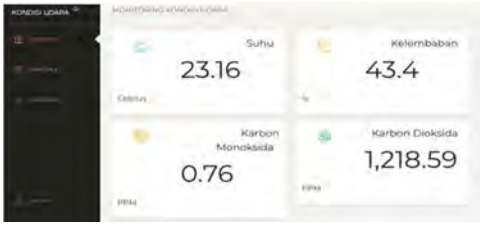

Fig 3. Dashboard Page

3. ***Table Page***, the table page, like Figure 4, displays the data taken from the sensor. On this page there are 10 data samples that are continuously updated along with the incoming data. The data displayed in the table list menu is the value of temperature, humidity, CO levels and CO2 levels as well as the time of data collection.

Fig 4. Table Page

4. ***Graph Page*,** if on the dashboard page the graph displayed is a combined graph of data on carbon monoxide and carbon dioxide levels, on the Graphic Data page, sensor data presented in graph form is displayed separately. The graphs are graphs of CO levels, CO2 levels, temperature values and humidity values. With this graph, it is expected to find out whether there is unreadable data on the website by looking at the time when the data entered the website.

Fig 5. Graph Page

D. *Decision Making*

Application of fuzzy logic method as artificial intelligence which is implemented in Waspmote Plug & Sense for environmental condition detection system. The reason for using the fuzzy logic method is because the concept of fuzzy logic is easy to understand and the mathematical concepts are fundamental in doing reasoning. Fuzzy logic can also be considered as an ambiguous or gray value that connects the input space to the output space.

The flow of the fuzzy logic decision making program begins with receiving data values from several sensors which later the data will be used as input for fuzzy logic. After the data is processed in fuzzy logic, data is generated in the form of values which will later be converted into environmental conditions. By applying the fuzzy logic method as a decision making system, before the decision making execution is carried out there are fuzzy stages in mapping the input space to the output space.

TABLE I. AIR QUALITY PARAMETERS

Category	Temperature (**ºC**)	Humidity (%RH)	CO (ppb/8 hours)	**CO_2** (ppb/8 hours)
Good	20-30	40-55	0-5	0-750
Average	31-40	56-65	6-10	751-1500
Bad	41-50	66-70	11-17	1501-2500
Dangerous	50+	70+	17+	2500+

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III. RESULTS AND DISCUSSION

A. *Testing Scenario*

The test is carried out in several stages. In each section will be shown the implementation and mechanism of data collection which will be tested from several samples.

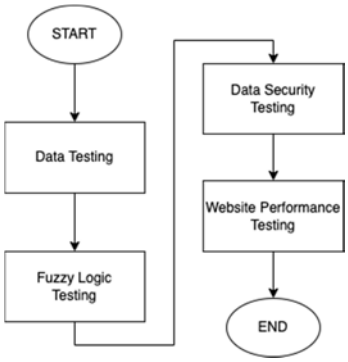

Fig 6. Testing Scenario

The location of data measurement and testing is carried out in the PENS campus area, namely in the D4 Building. The location was chosen because there are indoor and outdoor spaces in it.

B. *Data Testing*

Sensor data testing is carried out to determine the accuracy of sensor readings and sensor reliability to changes in environmental conditions. The test is done by comparing the sensor with HTC-1 digital at the same time. When testing the accuracy of sensor data, 3 different tests are carried out, this is done so that the value received by the sensor can change drastically.

1. *Indoor Testing*

Tests in an air-conditioned room were carried out to obtain a low temperature value and a low humidity value. This is done to determine changes in the temperature and humidity values detected by the sensor. From the tests carried out, data were obtained for temperature and humidity.

Test data were taken every five minutes. From the results obtained, it can be seen that the temperature reading from the Waspmote Plug & Sense temperature sensor for indoor ranges from 23°C to 24°C with different values ranging from 0.4°C to 0.6°C. Indoor Temperature Graph can be seen in Figure 7.

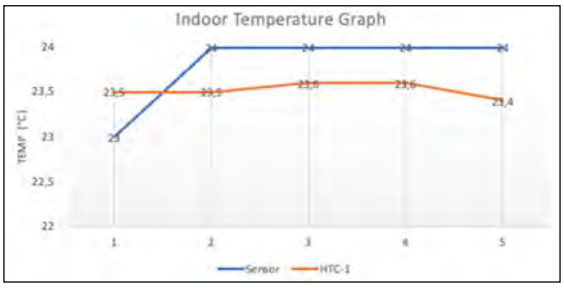

Fig 7. Indoor Temperature Graph

For humidity sensor readings the value ranges from 42% to 43% with a difference in value ranging from 1% to 4%. Indoor Humidity Graph can be seen in Figure 8.

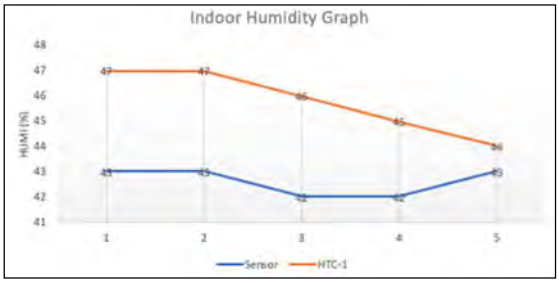

Fig 8. Indoor Humidity Graph

2. *Outdoor Testing.*

Outdoor testing is carried out to obtain high temperature values and high humidity values. This is done to determine changes in the temperature and humidity values detected by the sensor. In this test, it shows the data collection environment in the room, namely in the campus Lab, while Table 5 shows the outdoor data collection environment, namely on the basketball court. Data collection was carried out for 30 minutes with 5 minute intervals.

Test data were taken every five minutes. From the results obtained, it can be seen that the temperature reading from the Waspmote Plug & Sense temperature sensor for indoor ranges from 30°C to 34°C with different values ranging from 0.2°C to 0.8°C. Indoor Temperature Graph can be seen in Figure 9.

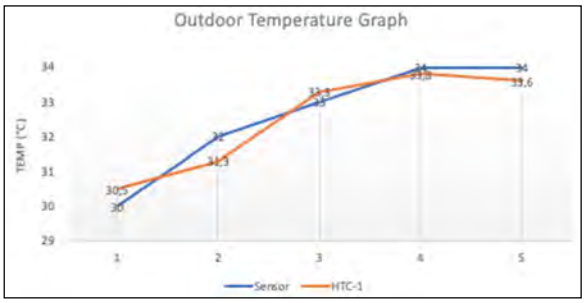

Fig 9. Outdoor Temperature Graph

For humidity sensor readings the value ranges from 52% to 64% with a difference in value ranging from 3% to 4%. Indoor Humidity Graph can be seen in Figure 10.

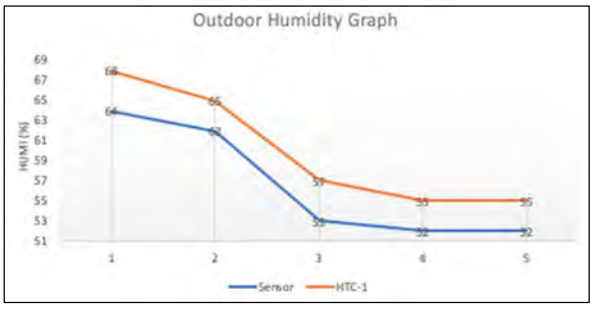

Figure 10. Outdoor Humidity Graph

Furthermore, from the results of testing sensor data both indoors and outdoors, we can compare the error presentation of each sensor.

The average error for the indoor temperature sensor is 1.86%, while the outdoor temperature sensor is 1.29%. The average error for indoor humidity is 3.52% and for outdoor it

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is 5.68%. The average error obtained proves that the Waspmote Plug & Sense sensor's ability to detect temperature and humidity is quite good.

C. *Fuzzy Logic Testing*

In the test, a comparison is made between the output of the fuzzy algorithm calculation in the program with the actual calculation of the fuzzy output.. With these rules, several variables can be combined to produce an output in the form of environmental health conditions. To be able to combine some of these rules, the "AND" operator is used in the fuzzy logic program.

Data	Program Result	Measurement Result	Error(%)
1	Good	Good	0
2	Good	Good	0
3	Good	Good	0
4	Good	Good	0
5	Good	Good	0
6	Average	Average	0
7	Average	Average	0
8	Good	Good	0
9	Average	Average	0
10	Good	Good	0
11	Good	Good	0
12	Good	Good	0
13	Good	Good	0
14	Good	Good	0
15	Good	Good	0

D. *Data Security Testing*

Data security uses the Advanced Encryption Standard (AES) algorithm as a data security method by encrypting fuzzy logic data, before the data is stored in the database. The data structure that will be encrypted is in the form of VARCHAR which is inputted automatically by Waspmote Plug & Sense.

1. *Encryption Testing*

AES algorithm is used to secure fuzzy logic data. The first test carried out on the encryption process is the measurement of the duration of time required during the encryption process.the results obtained are the average of the duration of each encrypted data, where each data is carried out 10 times testing the encryption duration. Data were taken from five different plaintext samples. Plaintext used is the result of Fuzzy Logic to determine environmental conditions.

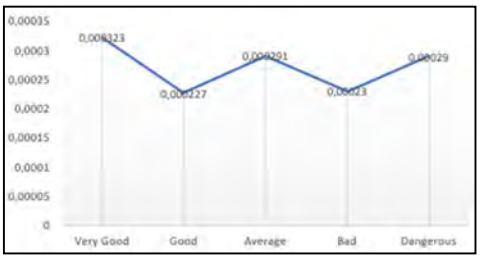

Fig 11. Encryption Duration Test Graph

From the Figure 11, it can be seen that the length of the character also affects the duration of encryption, although it is not too significant. Where the data contains "Very Good" with a length of 9 characters, has the longest average encryption duration of 0.000323s, while the data contains "Good" with a length of 4 characters, has the fastest average encryption duration of 0.000227s.

2. *Decryption Testing*

The next test carried out is measuring the duration of time required during the data decryption process. The results obtained are the average of the duration of each decrypted data, where each data is carried out 10 times for testing the duration of the decryption. Data were taken from five different ciphertext samples. The ciphertext used is the result of encryption from Fuzzy Logic data.

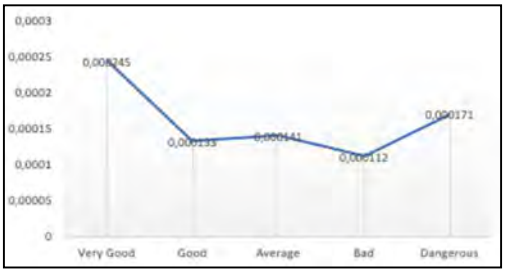

Fig 12. Decryption Duration Test Graph

From the Figure 12, it can be seen that the length of the character also affects the duration of the encryption, although it is not too significant. Where the data contains "Very Good" with a length of 9 characters, has the longest average duration of decryption of 0.000245s, while the data contains "Bad" with a length of 3 characters, has the fastest average duration of decryption of 0.000112s.

E. *Data Transmission Testing*

This test is carried out by taking data for Meshlium's position in Lab E-109 then Waspmote Plug & Sense moves away from the Lab to the D4 Basketball court. The monitoring process is carried out by sending data from Waspmote Plug & Sense which will then be seen whether the data will be received.

1. *Delay Testing*

Delivery time is the time it takes data to travel the distance from origin to destination. Delays in delivery time can be affected by distance, physical media, congestion or also long processing times. Delay itself is the accumulation of delay time from end to end on the Internet network. Delay is one of several parameters of Quality of Service (QoS).

$$Delay = Package\ Received\ Time - Packet\ Transfer\ Time \quad (1)$$

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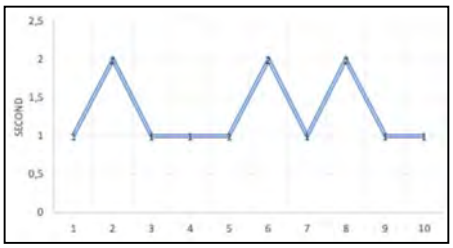

Fig 13. Delay Testing Graph

Based on the Figure 13, it can be seen that the delay generated by each time condition has a different value but tends to be the same. This is caused by several factors, namely the distance between the sender and receiver, physical media and the long process of the network. From the data obtained, it can be seen that the average delay generated is 1.3second.

2. *Throughput Testing*

Throughput is the average speed of data received by a node in a certain observation time interval. Throughput is the actual bandwidth at that time where we are transferring data. It has the same unit as bandwidth, namely bytes per second (bps).

$$Throughput = \frac{Amount\ of\ data\ sent}{Packet\ Transfer\ Time} \quad (2)$$

In this test used 10 test samples. The observed data are the number of bits sent and the delay, then both are entered into the formula and displayed as the test result.

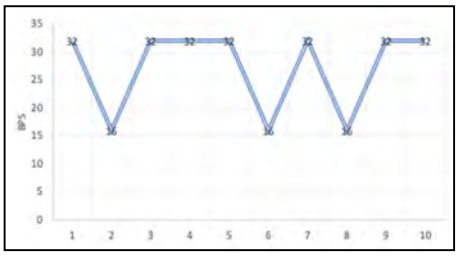

Fig 14. Throughtput Testing Graph

Based on the Figure 14, it can be seen that the Throughput generated at each time condition has a different value but tends to be the same. Things that affect the speed of delivery throughput are network devices, types of data transferred, network traffic used, client computer specifications, server specifications. From the data obtained, it can be seen that the average throughput generated is 27.2 bps.

F. *Website Performance Testing*

This test is carried out to determine the speed of the website in loading all html-php programming files so that they can be displayed on a website page and can be read by users.

This test is carried out using 4 samples of loading time, namely in the morning (at 07:00 WIB), noon (at 13:00 WIB), afternoon (at 16:00 WIB) and evening (at 21 : 00 WIB). Each time sample there are 5 times of data collection for each page, namely the dashboard, table data and graph data, then the data is averaged. Here are the results of the website loading test.

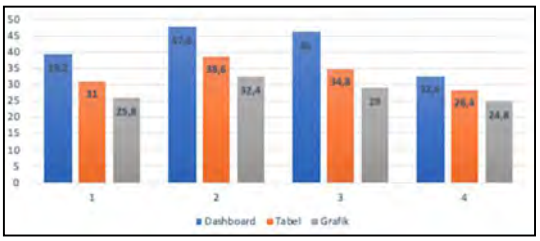

Fig 15. Loading Time Graph

In Figure 15, information can be obtained that the longest address retrieval is during the day or evening. Whereas in the morning and evening it takes a little time to load the website. This can be caused by factors such as the internet network, the device and browser used or the quality of the server.

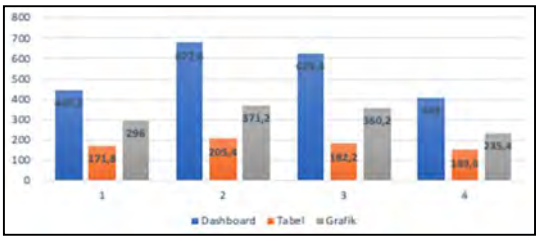

Fig 16. Scripting Time Graph

On the dashboard page, there is a script to call several different programming files to be able to display the current CO and CO2 values and an overall graph. Because of that on the web page it takes the longest script retrieval time of up to 691.6 ms during the day.

IV. CONCLUSION

Based on the results of system processing that has been carried out in this study, several conclusions were obtained as follows. The system created is able to help users to find out the air condition in the surrounding environment in real time. The website as a place to display data from sensors stored in the database can also run well and in a short time. Fuzzy logic which functions as a decision-making algorithm for environmental conditions can also work properly. The calculation of the fuzzy logic algorithm through the program is in accordance with the actual mathematical calculations.

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